

PAPER • OPEN ACCESS

## Predicting the Compressive Strength of Concrete by using Machine Learning Techniques

To cite this article: Mary Devika Bandaru *et al* 2024 *J. Phys.: Conf. Ser.* **2779** 012016

View the [article online](#) for updates and enhancements.

You may also like

- [Preparation and regulation of electromagnetic parameters of  \$Ti\_3C\_2T\_x@Ni\$  composite powders](#)  
Shujun Hu, Shibo Li, Peng Pan et al.
- [Sustainable Development of 3D-Printable Quaternary Geopolymer Concrete](#)  
B Charan Kumar Reddy, Sk Khaja Sameer, V Ramesh et al.
- [Membership Study of Open Cluster NGC 6134 using HDBSCAN Clustering Algorithm](#)  
M Yusuf, D G Ramadhan, A T P Jatmiko et al.



**PRIME<sup>TM</sup>**  
PACIFIC RIM MEETING  
ON ELECTROCHEMICAL  
AND SOLID STATE SCIENCE  
**HONOLULU, HI**  
October 6-11, 2024

*Joint International Meeting of*  
The Electrochemical Society of Japan (ECSJ)  
The Korean Electrochemical Society (KECS)  
The Electrochemical Society (ECS)

Early Registration Deadline:  
**September 3, 2024**

**MAKE YOUR PLANS  
NOW!**

# PREDICTING THE COMPRESSIVE STRENGTH OF CONCRETE BY USING MACHINE LEARNING TECHNIQUES

Mary Devika Bandaru<sup>1</sup>, Suseela Kyle<sup>1</sup>, Tallapudi Indira Priyadarshini<sup>1</sup>,  
Durga Vara Prasad Bokka<sup>1</sup> & Ungarala Ganesh Sai<sup>2</sup>

<sup>1</sup> Assistant professor, Department of Civil Engineering, Vishnu Institute of Technology, Vishnupur, Bhimavaram & 534202, India.

<sup>2</sup> Student, Department of Civil Engineering, Vishnu Institute of Technology, Vishnupur, Bhimavaram & 534202, India.

Corresponding Author Email ID: [20pa1a0155@vishnu.edu.in](mailto:20pa1a0155@vishnu.edu.in)

**Abstract.** Considerable efforts have been made to increase the strength of concrete by substituting some of the cement in the concrete with industrial waste, such as fly ash. Predicting the compressive strength of concrete, however, is a difficult undertaking since it depends on a number of elements, including the water-to-cement ratio and the size and form of the particles. The work on machine learning algorithms for Evaluating the strength of concrete with the inclusion of fly ash is presented in this publication. In order to determine the most accurate estimation technique, the study also compares the accuracy of various machine learning models, including ensemble models and non-ensemble models in predicting CS. For this purpose, a dataset with a diverse set of experimental findings, encompassing a broad spectrum of compressive strength values, was gathered from established literature and verified through statistical scrutiny. The amounts of cement, fine aggregate, coarse aggregates, fly ash, water content, percentage of superplasticizer, and curing days were the main input parameters for the ML models, while the output was the CS of concrete. The assessment of performance involved the utilization of various performance metrics, including MSE & R2 to assess accuracy and reliability. The comparison reveals that XGBoost Regressor, Bagging Regressor, and Random Forest are the most reliable models. Fly ash and other parameters' effects on CS prediction were also fully understood thanks to the use of sensitivity and parametric analysis, which also helped to shed light on the relationship between input parameters and CS. The efficiency of machine learning methods is demonstrated in the study. It eliminates the need for expensive and time-consuming experimental experiments by providing researchers with a quicker and more economical way to assess how fly ash and other factors affect CS estimation.

## 1. Introduction

Various ashes, including fly ash (FA) and ground granulated blast furnace slag (GGBS), are used in geopolymer concrete (GPC). While FA is commonly used, its reliance on heat curing limits site project applications. GGBS is explored as a FA replacement to address this issue. Machine learning models (ANN, M5P, LR, MLR) are employed to predict compressive strength of GGBS/FA-GPC, considering eleven input parameters. The ANN model outperforms others, with alkaline liquid to binder ratio, FA content, molarity, and sodium



silicate content being significant factors [1]. Geopolymer concrete (GPC) is advocated to minimize environmental harm from cement production. Predicting its compressive strength (C–S) is vital for design. Machine learning models (ANNs, DNNs, ResNet) are developed using extensive datasets, with ResNet showing superior predictive capability. Curing temperature, alkaline liquid to binder ratio, and sodium silicate amount are crucial predictors [2]. Adding fly ash improves physical and mechanical properties of concrete. Tests show enhanced tightness and resistance to chloride ion penetration. European standards may need amendment to allow fly ash in concrete with specific cement classes [3]. A new optimization algorithm, Cat Swarm Optimization (CSO), is proposed, inspired by cat behavior. Experiments show CSO outperforms Particle Swarm Optimization (PSO) [4].

Machine learning approaches are used to predict high-strength concrete (HSC) behavior, offering accuracy and efficiency over traditional methods. Deep learning proves effective in material quantity estimation [5]. Gene Expression Programming (GEP) is employed to predict concrete strength under true-triaxial loading, offering a faster alternative to lab experiments [6]. Advanced machine learning techniques predict compressive strength of rice husk ash blended concrete (RHAC) more accurately than traditional methods, reducing laboratory experiment costs [7].

Soft computing techniques, namely Random Forest Regression (RFR) and Gene Expression Programming (GEP), are used to predict compressive strength of FA-based geopolymer concrete (FGPC), offering reliable alternatives to experimental procedures [8]. GEP is used to predict concrete strength under triaxial compression loading, offering a robust model for design purposes [9]. Artificial Neural Networks (ANN) are adapted to predict compressive strength of high-performance concrete (HPC), demonstrating superior accuracy over regression analysis [10]. A hybrid model combining Firefly Algorithm (FA) and Least Squares Support Vector Regression (LS-SVR) predicts HPC compressive strength effectively [11].

Multiple Regression Analysis (MRA) and ANN models predict compressive strength of High Performance Concrete containing nano silica and copper slag, with ANN showing higher accuracy [12]. Ensemble methods like Adaboost can outperform single classifiers, offering better prediction in various applications [13]. Multi-layer feed-forward neural networks (MFNNs) effectively predict concrete compressive strength, offering practical benefits in modeling complex relationships [14].

XGBoost, a tree boosting system, provides efficient and scalable solutions for machine learning tasks, particularly on large datasets [15]. Principal component analysis and radial basisfunction neural networks are employed to predict concrete compressive strength, achieving high accuracy and meeting engineering test accuracy requirements [16]. Deep learning techniques, such as neural networks, offer efficient design solutions for prestressed concrete members, improving structural engineering practices [17]. Adaptive Neuro-Fuzzy Inference System (ANFIS) predicts compressive strength of self-compacting concrete containing treated palm oil fuel ash, demonstrating high accuracy and efficiency over experimental methods [18].

## **2. Overview of Machine Learning Algorithms**

Machine learning models, drawing inspiration from the structural principles of the human brain, undergo training processes on diverse datasets, enabling them to compute, adapt, and predict new data. Comprising input and output layers with corresponding variables, these models incorporate hidden layers that execute essential calculations, revealing concealed patterns and relationships within the input datasets.

### *2.1. Non-ensemble Model*

A non-ensemble model, also known as a standalone model, refers to a machine-learning model that is trained and operates independently without being combined with other models to form an ensemble. In machine learning, in general, blending the forecasts of various models is a

common practice to enhance overall effectiveness compared to relying solely on individual model outputs. These models are single, self-contained entities that make predictions based on the features and patterns they have learned during the training process and can belong to various types, such as linear regression, support vector machines, decision trees and more. A non-ensemble model relies solely on its own internal parameters and structure.

*2.1.1. Support Vector Regressor.* Support Vector Regression (S.V.R.) is a non-ensemble learning model used for regression tasks. Unlike traditional regression models, S.V.R. aims to find a hyperplane that best captures the relationships between input variables and their corresponding continuous target values. It makes use of crucial data points known as support vectors to define the hyperplane and a margin that allows for flexibility in accommodating variations in the data. S.V.R. excels in managing nonlinear associations through the use of kernel functions that transform input data into a higher-dimensional realm. By optimizing a cost function that balances the fit to the training data and the smoothness of the solution, S.V.R. provides a robust approach to regression tasks, especially in scenarios with complex and nonlinear non-linear patterns.

$$f(x, w) = \sum_{i=1}^n \omega_i g_1(x) + b$$

*2.1.2. K.N.N. Regressor.* K-Nearest Neighbours (K.N.N.) Regressor is employed in regression tasks, making predictions for the numerical value of a target variable through the average or weighted average of its k-nearest neighbours in the feature space. In this context, "k" represents the number of neighbouring data points considered for prediction. The distances between the input data point and all other data points in the training set are computed by the algorithm, and the k-nearest neighbors are then chosen based on these calculations and computes their average (or weighted average) as the predicted value for the target variable. K.N.N. Regressor is a simple yet effective algorithm for regression tasks, especially when dealing with relatively small datasets and in situations where the underlying relationships are locally defined.

*2.1.3. Decision Tree Regressor.* Decision Tree Regressor, which operates by recursively dividing the input dataset into subsets according to feature values, leading to the formation of a tree-like structure where each internal node signifies a decision based on a particular feature, leading to subsequent branches and leaf nodes containing predicted continuous values. The model is trained to minimize the variance within each leaf node, optimizing for accurate predictions of the target variable. Decision Tree Regressors are versatile and interpretable, making them suitable for various regression tasks, though they may be prone to overfitting with complex structures.

## *2.2. Ensemble models*

Creating an amalgamation of predictions from various individual models, ensemble techniques in machine learning aim to enhance accuracy and robustness by leveraging the diversity of these models, often surpassing the performance of any singular model, especially when the individual models have different strengths and weaknesses.

*2.2.1. Random Forest Regressor.* The Random Forest, a robust ensemble learning model widely employed in machine learning, distinguishes itself from non-ensemble counterparts by harnessing the collective power of numerous decision trees. These trees are individually crafted using random subsets of both training data and features, and their predictions are amalgamated through a voting mechanism. This strategy not only fosters a more accurate overall prediction but also addresses overfitting concerns, bolstering the model's ability to generalize effectively.

Consequently, Random Forest stands as a favoured option for classification and regression tasks across diverse domains such as finance, healthcare, and image recognition.

**2.2.2. Gradient Boost Regressor.** The Gradient Boosting Regressor stands out as a formidable non-ensemble learning algorithm within the boosting family. Diverging from conventional non-ensemble models, it constructs a potent predictive framework through successive training of weak learners, typically decision trees, and fine-tuning them to rectify errors introduced by the preceding models. By iteratively incorporating new models, each dedicated to addressing residual errors in the collective ensemble, it systematically minimizes a loss function. This stepwise refinement enables the model to progressively enhance its predictive capabilities and adeptly handle intricate relationships within the dataset. Renowned for its exceptional predictive accuracy and resilience, the Gradient Boosting Regressor remains a favoured option for a diverse range of regression tasks.

**2.2.3. Ada Boost Regressor.** Ada Boost Regressor, diverging from conventional non-ensemble regression models, constructs a potent predictive framework for regression tasks by amalgamating forecasts from various weak learners, commonly decision trees. Ada Boost Regressor sequentially trains a series of weak learners, assigning higher weights to the instances that were inaccurately predicted in the previous iteration. This adaptive weighting mechanism enables Ada Boost Regressor to focus on the challenging data points, improving overall model performance. The ultimate forecast is derived through a weighted combination of predictions from individual weak learners, where the contribution of each weak learner is scaled based on its accuracy. Ada Boost Regressor is recognized for its capability to address intricate data relationships and alleviate overfitting, establishing itself as a potent solution for regression challenges.

**2.2.4. Bagging Regressor.** The Bagging Regressor, an acronym for Bootstrap Aggregating, is a regression model employed in machine learning for regression assignments. Its methodology involves training numerous copies of the identical regression algorithm on distinct bootstrap samples—random subsets of the initial training dataset with replacement. The ultimate prediction is derived by averaging the forecasts generated by each unique model, thereby reducing variance and enhancing the overall robustness of the regression model. By leveraging the diversity of the bootstrap samples, the Bagging Regressor mitigates overfitting and increases predictive accuracy, making it a valuable tool in non-ensemble settings for improving the stability and performance of regression algorithms. The choice between using a non-ensemble or an ensemble model depends on the specific problem, dataset characteristics, and performance requirements. Non-ensemble models are simpler and computationally less intensive, making them suitable for certain tasks where interpretability and efficiency are crucial.

### 2.3. Performance parameters

To enhance the assessment of predictive model performance, two distinct metrics are incorporated, each having its own defined criteria.

Mean Squared Error (M.S.E.)

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

Coefficient of determination R-Square ( $R^2$ )

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2}$$

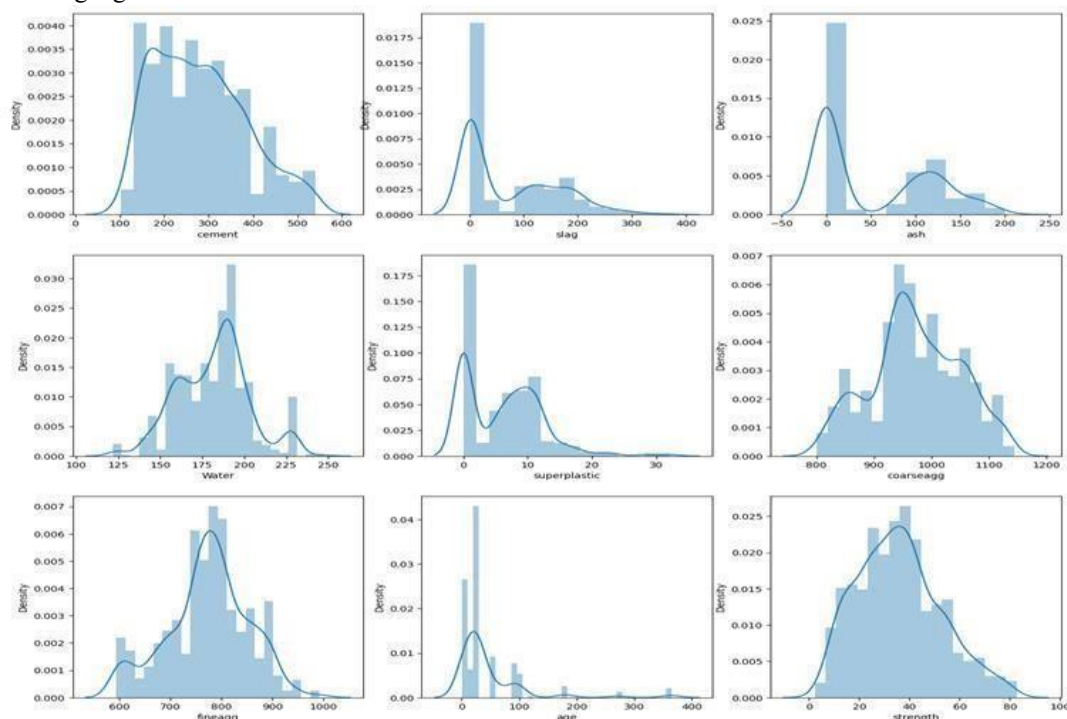
Here  $\hat{y}$  = Predicted value of  $y$   
 $\bar{y}_i$  = Average value of  $y$   
 $\bar{y}$  = Mean value of all the calculated values  
 $N$  = Total no of samples in the data set

### 3. Description of dataset

As stated earlier, this investigation conducted experiments to evaluate the compressive strength of concrete through machine learning algorithms, utilizing a dataset derived from the research conducted by Yeh [22], which included 1030 stress versus compression tests. The dataset comprises eight input variables:

- Cement.
- Fine aggregate.
- Fly ash.
- Blast furnace slag.
- Superplasticizer.
- Water.
- Coarse aggregate.
- Age.

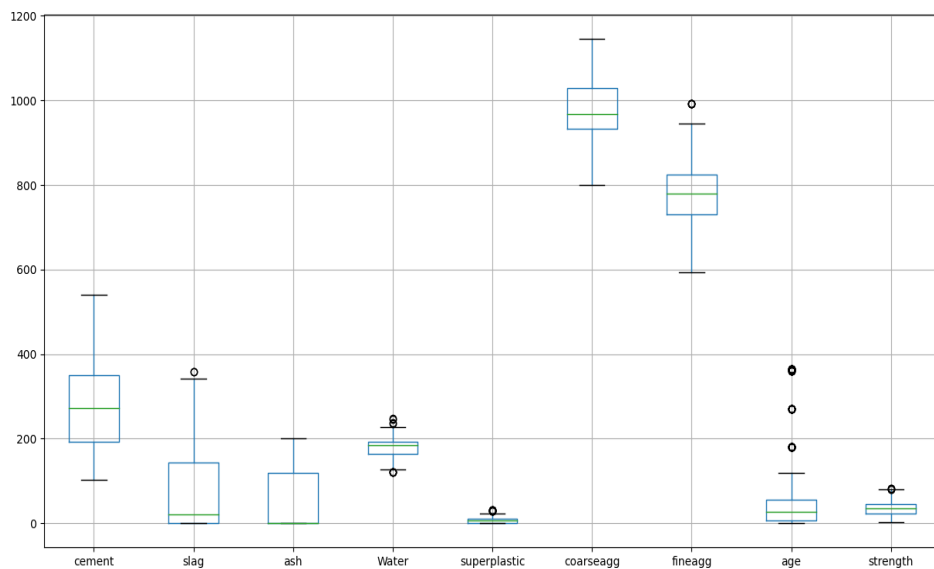
The compressive strength of concrete ( $F_c$ ) serves as the dependent variable in this study. Below, figure1 displays visual representations of histograms and density boxes generated from the employed dataset. As mentioned earlier, the aim of these visual aids is to improve comprehension of the appropriateness of various techniques for achieving optimal compatibility with machine learning models. The variables showcase distributions that are closely resembling normal distributions, suggesting the potential enhancement of efficiency in learning algorithms.



**Figure 1.** Histograms and density plots



In figure 2 presented beneath, box plots illustrate the variables being examined. Furthermore, an evaluation of outlier values was carried out for both input and output variables, with the results documented in Table 1. The examination indicates a uniformly distributed dataset, with none of the attributes displaying over 10% outliers.



**Figure 2.** Box plot

The correlation matrix illustrated in Figure 3 for the utilized dataset indicates that the variables



**Figure 3.** Correlation matrix

within the computational model display a notable absence of correlation, suggesting a primarily independent relationship among the components.

In the formulation of the predictive models for this investigation, the experimental data obtained was segregated into two separate sets: the training set, pivotal in shaping the ultimate resilient learner, and the testing set, utilized for evaluating the model's precision in forecasting compressive strength. The training phase involved 85% of the data (875 samples), leaving the remaining 15% (155 samples) exclusively for testing in this research. It is noteworthy that the selection of optimal parameters for each method was facilitated by cross-validation, eliminating the necessity for a separate data validation set.

#### 4. Statistical analysis and K-fold cross validation study

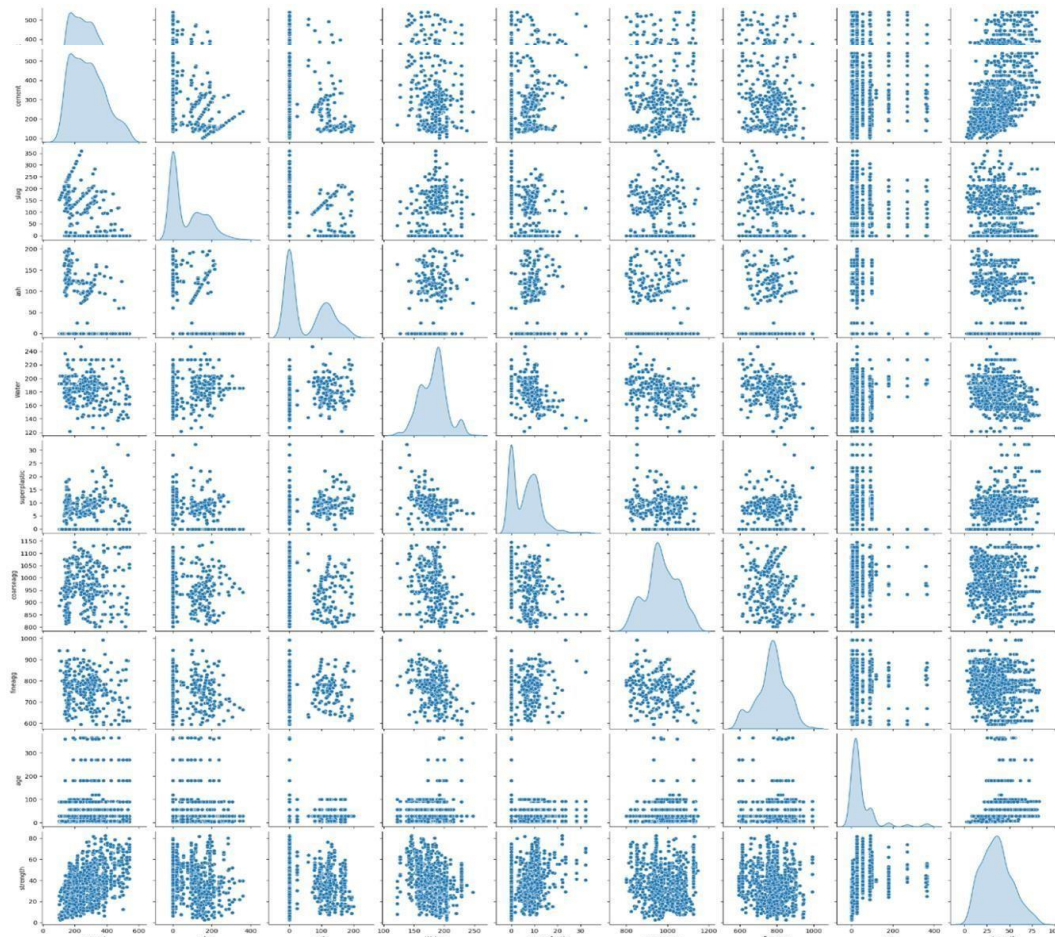
Statistical analysis and K-fold cross-validation are essential components in evaluating the performance and robustness of predictive models in machine learning. Statistical analysis involves examining the relationships within the data, ensuring that the chosen model captures the underlying characteristics effectively. This analysis often includes measures such as mean, median, standard deviation, and correlation coefficients to gain insights into the distribution and correlations among variables.

K-fold cross-validation, On the contrary, cross-validation serves as a method to evaluate the model's generalization capability and counteract overfitting. The dataset undergoes division into k subsets or folds, and the model undergoes training and validation k times, utilizing a distinct fold for validation in each iteration while employing the remaining folds for training. This iterative process aids in estimating the model's performance on unseen data, offering a more robust evaluation in comparison to a singular train-test split. When combined, statistical analysis and K-fold cross-validation contribute to a comprehensive assessment of the model's effectiveness., ensuring its reliability, generalization capability, and suitability for real-world applications. The implementation of these methodologies is essential to develop models that exhibit strong performance not only on the training dataset but also demonstrate effective generalization to unfamiliar data.

**Table 1.** Statistical indication of parameters

| Ingredients      | Count  | Mean    | Std     | Min     | 25%     | 50%     | 75%          | Max      |
|------------------|--------|---------|---------|---------|---------|---------|--------------|----------|
| Cement           | 1030.0 | 281.167 | 104.506 | 102.00  | 192.375 | 272.900 | 350.000      | 540.000  |
| Slag (GGBS)      | 1030.0 | 73.895  | 86.279  | 0.000   | 0.000   | 22.000  | 142.950      | 359.400  |
| Fly Ash          | 1030.0 | 54.188  | 63.997  | 0.000   | 0.000   | 0.000   | 118.300      | 200.100  |
| Water            | 1030.0 | 181.567 | 21.354  | 121.800 | 164.900 | 185.000 | 192.000      | 247.000  |
| Superplasticizer | 1030.0 | 6.204   | 5.973   | 0.000   | 0.000   | 6.400   | 10.200       | 32.200   |
| Coarse Aggregate | 1030.0 | 972.918 | 77.753  | 801.00  | 932.000 | 968.000 | 1029.40<br>0 | 1145.000 |
| Fine Aggregate   | 1030.0 | 773.580 | 80.175  | 954.000 | 730.950 | 779.500 | 824.000      | 992.600  |
| Age              | 1030.0 | 45.662  | 63.169  | 1.000   | 7.000   | 28.000  | 56.000       | 365.000  |
| Strength         | 1030.0 | 35.817  | 16.705  | 2.330   | 23.710  | 34.445  | 46.135       | 82.600   |





**Figure 4.** Relationship of input variables with compressive strength

A model demonstrates reliability and significance when its  $R^2$  value surpasses 0.8, with a closer approach to 1 signifying increased precision in predictions. The M.S.E. and RMSE metrics highlight the disparity between tested and predicted results, where heightened prediction accuracy corresponds to diminished M.S.E. and RMSE values. Additionally, a model's improved performance is reflected in lower M.A.E. values, approaching zero, as M.A.E. signifies the actual error in predictions. Furthermore, the MAPE metric unveils the ratio of prediction error to tested values.

## 5. Results and discussion of the model

### 5.1. Results

**5.1.1. Support Vector Regressor.** The evaluation outcomes derived from tests and predictions employing the S.V.R. model indicated diminished accuracy, characterized by an  $R^2 = 0.65$  and  $M.S.E = 90.42$ . The utilization of S.V.R. revealed an average model performance. These approaches seek to pinpoint the most effective combination of hyperparameters to enhance the model's overall performance. In the current study, the S.V.R. model accuracy before k-fold validation is 0.65, and after k-fold validation, it is 0.69.

**5.1.2. KNN Regressor.** The comparative analysis of test and prediction results using the K.N.N. Regressor model revealed reduced accuracy, evidenced by an  $R^2 = 0.75$  &  $M.S.E = 64.43$ . It

was noted that employing the K.N.N. Regressor led to an average model performance. The objective of employing these methods is to pinpoint the most effective combination of hyperparameters that result in the optimal performance of the model. In the current study, the K.N.N. Regressor model accuracy before k-fold validation is 0.75, and after k-fold validation, it is 0.71.

*5.1.3. Decision Tree Regressor.* The evaluation of test and prediction results using the Decision Tree Regressor model demonstrated decreased accuracy, indicated by an adjusted  $R^2 = 0.75$  &  $M.S.E = 116.16$ . It was observed that the utilization of the Decision Tree Regressor led to an average model performance. The main objective of employing these techniques is to pinpoint the most effective combination of hyperparameters that enhances the model's overall performance. In the current study, the Decision Tree Regressor model accuracy before k-fold validation is 0.75, and after k-fold validation, it is 0.85.

*5.1.4. Random Forest Regressor.* The analysis of test and prediction outcomes using the Random Forest (R.F.) model revealed a decrease in accuracy, indicated by an adjusted  $R^2 = 0.87$  and  $M.S.E = 31.85$ . It was noted that employing the Random Forest model led to an average performance. The goal of employing these methods is to pinpoint the most effective combination of hyperparameters that enhances the model's overall performance. In the current study, the R.F. model accuracy before k-fold validation is 0.87 and after k-fold validation is 0.90.

*5.1.5. Gradient Boost Regressor.* The evaluation of test and prediction results using the Gradient Boosting Regressor (G.B.R.) model revealed a decrease in accuracy, indicated by an  $R^2 = 0.88$  and  $M.S.E = 31.41$ . It was observed that employing the G.B.R. model led to an average performance. The main objective of employing these methods is to pinpoint the most effective combination of hyperparameters that enhances the model's overall performance. In the current study, the G.B.R. model accuracy before k-fold validation is 0.88 and after k-fold validation is 0.89.

*5.1.6. AdaBoost Regressor.* The assessment of test and prediction outcomes employing the AdaBoost Regressor model showed reduced accuracy, as evidenced by an  $R^2 = 0.76$  &  $M.S.E = 66.07$ . It was noted that the application of the AdaBoost Regressor resulted in an average model performance. The goal of these techniques is to identify the optimal set of hyperparameters that maximizes the model's performance. In the current study, the AdaBoost Regressor model accuracy before k-fold validation is 0.76, and after k-fold validation, it is 0.78.

*5.1.7. Bagging Regressor.* The evaluation of test and prediction outcomes using the Bagging Regressor model revealed a decrease in accuracy, as indicated by an adjusted  $R^2$  value of 0.85 and  $M.S.E.$  of 38.69. It was observed that utilizing the Bagging Regressor resulted in an average model performance. The primary aim of these techniques is to identify the optimal set of hyperparameters that maximizes the model's performance. In the current study, the Bagging regressor model accuracy before k-fold validation is 0.85 and after k-fold validation, is 0.89.

**Table 2.** Comparison of studied ML algorithms

| Algorithm     | Accuracy |
|---------------|----------|
| Random Forest | 0.878448 |
| R-F K-fold    | 0.908329 |

|                                 |          |
|---------------------------------|----------|
| Gradient Boost Regressor        | 0.880136 |
| Gradient Boost Regressor K-fold | 0.897393 |
| AdaBoost Regressor              | 0.766126 |
| AdaBoost Regressor K-fold       | 0.781754 |
| KNN Regressor                   | 0.754171 |
| KNN Regressor K-fold            | 0.705797 |
| Bagging Regressor               | 0.852356 |
| Bagging Regressor K-fold        | 0.891669 |
| SVR                             | 0.654996 |
| SVR K-fold                      | 0.697300 |
| DTR-1                           | 0.759363 |
| DTR K-fold                      | 0.852336 |
| DTR – 2                         | 0.730958 |
| Pruned Decision Tree            | 0.556821 |

**5.2 Model comparison.** Distinct combinations of cement, fine aggregate (F.A.), coarse aggregates (C.A.), fly ash (F.A.), water content, percentage of superplasticizer, and curing days lead to varied compressive strengths (C.S.) in concrete. It is widely acknowledged that materials exhibit diverse compositions corresponding to their compressive strength ranges. Consequently, Table 2 delineates the model's efficacy across various strength categories. From the data in Table 2, it was evident that both the Random Forest Regressor k-fold model and the Bagging Regressor k-fold model demonstrated superior performance, especially when forecasting concrete compressive strength (C.S.), the model demonstrated notable accuracy with R2 values of 0.87 and 0.85, particularly emphasizing its precision in the 6-31 MPa range due to the higher density of data points within this specific interval.

## 6. Conclusion

This study aims to explore the application of computational intelligence in determining concrete compressive strength using a database sourced from Yeh's studies. Three machine learning models specifically, Random Forest, Support Vector Machines, and Bagging Regression were employed. Data pre-processing techniques and visualization methods were also utilized to enhance the outcomes. Cross-validation, a pre-processing method, was employed to enhance the performance of machine learning models. The results obtained indicate that the Random Forest model exhibited the best performance (M.S.E. = 31.85 and  $R^2 = 0.87$ ) with an average execution time. Notably, the Support Vector Machine (SVM) method had the least favourable performance but demonstrated the shortest execution time. Conversely, the Bagging Regressor yielded the second-best result but required the longest execution time. The overall error rate was found to be low, suggesting that these techniques can effectively predict concrete compressive strength within the acceptable safety range for engineering practices. Computational intelligence models, showcased in this study, prove to be reliable for addressing various complex problems, including prediction challenges. They can serve as valuable alternatives when deviations in available data are expected and accepted, and when a defined methodology is unavailable. Therefore, in the pursuit of predicting concrete properties with high reliability, computational intelligence models offer a viable and cost-effective alternative to traditional and innovative models, replacing the need for expensive experimental investigations. In addition, the study delves into the influence of diverse

compositions of cement, fine aggregate (F.A.), coarse aggregates (C.A.), fly ash (F.A.), water content, percentage of superplasticizer, and curing days on concrete compressive strength. Subsequent research aimed at evaluating model performance across different strength ranges, as outlined in Table 2, revealed that the Random Forest Regressor k-fold model and Bagging Regressor k-fold model demonstrated superior performance, particularly in predicting concrete compressive strength, with  $R^2$  values of 0.87 and 0.85, respectively. Notably, the model's accuracy was more pronounced in the 6–31 MPa range due to the larger size of data points within this specific range.

## 7. Reference

- [1] H. U. Ahmed, A. A. Mohammed, and A. Mohammed, "Soft computing models to predict the compressive strength of GGBS/FAgeopolymer concrete," *PLoS ONE*, vol. 17, no. 5, May 2022, doi: 10.1371/journal.pone.0265846.
- [2] D. A. Emarah, "Compressive strength analysis of fly ash-based geopolymer concrete using machine learning approaches," *Results in Materials*, vol. 16, Dec. 2022, doi: 10.1016/j.rinma.2022.100347.
- [3] A. Szcześniak, J. Zychowicz, and A. Stolarski, "Influence of fly ash additive on the properties of concrete with slag cement," *Materials*, vol. 13, no. 15, 2020, doi: 10.3390/MA13153265.
- [4] S.-C. Chu and P.-W. Tsai, "Information and Control ICIC International c °2007 ISSN," 2007. [Online]. Available: <https://www.researchgate.net/publication/228721750>
- [5] F. Aslam et al., "Applications of Gene Expression Programming for Estimating Compressive Strength of High-Strength Concrete," *Advances in Civil Engineering*, vol. 2020, 2020, doi: 10.1155/2020/8850535.
- [6] S. K. Babanajad, A. H. Gandomi, and A. H. Alavi, "New prediction models for concrete ultimate strength under true-triaxial stress states: An evolutionary approach," *Advances in Engineering Software*, vol. 110, pp. 55–68, Aug. 2017, doi: 10.1016/j.advengsoft.2017.03.011.
- [7] M. N. Amin, A. Iqtidar, K. Khan, M. F. Javed, F. I. Shalabi, and M. G. Qadir, "Comparison of machine learning approaches with traditional methods for predicting the compressive strength of rice husk ash concrete," *Crystals*, vol. 11, no. 7, Jul. 2021, doi: 10.3390/cryst11070779.
- [8] M. A. Khan, S. A. Memon, F. Farooq, M. F. Javed, F. Aslam, and R. Alyousef, "Compressive Strength of Fly-Ash-Based Geopolymer Concrete by Gene Expression Programming and Random Forest," *Advances in Civil Engineering*, vol. 2021, 2021, doi: 10.1155/2021/6618407.
- [9] A. H. Gandomi, S. K. Babanajad, A. H. Alavi, and Y. Farnam, "Novel Approach to Strength Modeling of Concrete under Triaxial Compression," *Journal of Materials in Civil Engineering*, vol. 24, no. 9, pp. 1132–1143, Sep. 2012, doi: 10.1061/(asce)mt.1943-5533.0000494.
- [10] Siddique, R., Aggarwal, P., Aggarwal, Y., "Prediction of compressive strength of self- compacting concrete containing bottom ash using artificial neural networks," *Adv. Eng. Softw.* 42 (2011) 780–786.
- [10] I. C. Yeh, "Modeling of strength of high-performance concrete using artificial neural networks," *Cement and Concrete Research*, vol. 28, no. 12, pp. 1797–1808, 1998, doi: 10.1016/S0008-8846(98)00165-3
- [11] A. D. Pham, N.-D. Hoang, and Q.-T. Nguyen, "Predicting Compressive Strength of High-Performance Concrete Using Metaheuristic-Optimized Least Squares Support Vector Regression," *Journal of Computing in Civil Engineering*, vol. 30, no. 3, May 2016, doi: 10.1061/(asce)cp.1943-5487.0000506.
- [12] S. Chithra, S. R. R. S. Kumar, K. Chinnaraju, and F. Alfin Ashmita, "A comparative study on the compressive strength prediction models for High Performance Concrete

- containing nano silica and copper slag using regression analysis and Artificial Neural Networks,” *Construction and Building Materials*, vol. 114, pp. 528–535, Jul. 2016, doi: 10.1016/j.conbuildmat.2016.03.214.
- [13] T. G. Dietterich, “Ensemble Methods in Machine Learning.” [Online]. Available: <http://www.cs.orst.edu/~tgdt>
- [14] N. Hong-Guang and J.-Z. Wang, "Prediction of compressive strength of concrete by neural networks." Chou, J. S., Tsai, C. F., Pham, A. D., Lu, Y. H., *Machine learning in concrete strength simulations: Multi-nation data analytics*, *Constr. Build. Mater.* 73 (2014) 771-780.
- [15] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, Association for Computing Machinery, Aug. 2016, pp. 785-794. doi: 10.1145/2939672.2939785.
- [16] Z. Sun, K. Li, and Z. Li, "Prediction of concrete compressive strength based on principal component analysis and radial basis function neural network," in *IOP Conference Series: Materials Science and Engineering*, IOP Publishing Ltd, Dec. 2019. doi: 10.1088/1757-899X/677/2/022045.
- [17] A. A. Torky and A. A. Aburawwash, "A Deep Learning Approach to Automated Structural Engineering of Prestressed Members," *International Journal of Structural and Civil Engineering Research*, pp. 347-352, 2018, doi: 10.18178/ijscer.7.4.347-352.
- [18] T. Al-Mughanam, T. H. H. Aldhyani, B. Alsubari, and M. Al-Yaari, “Modeling of compressive strength of sustainable self-compacting concrete incorporating treated palm oil fuel ash using artificial neural network,” *Sustainability (Switzerland)*, vol. 12, no. 22, pp. 1–13, Nov. 2020, doi: 10.3390/su12229322.